Akoizo Original Practice - Form 2 - Answer key

Division B | Thermodynamics | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Radiation requires matter between objects?

Unit: Conduction, convection, radiation, and heat capacity.
Worked-case data: Heat 0.20 kg water by 10 K with c=4180 J/(kg K), neglecting losses.

A. Only in a vacuum

B. Only for visible light

C. No

D. Always

Correct: C

2. (2 points) At a phase transition, use…

Unit: Latent heat, heating curves, and calorimetry.
Worked-case data: Melt 0.10 kg ice at 0 °C using Lf=334 kJ/kg. Find energy.

A. Only mcdelta T with zero delta T

B. Only PV

C. Only temperature ratio

D. mL

Correct: D

3. (2 points) What is the internal-energy change?

Unit: Gas processes with graphs and simple algebra.
Worked-case data: Pressure: 100,000 Pa, held constant. Volume change: An increase of 0.002 m^3. Heat input: 300 J enters the gas.
Separate investigation observations:
Pressure: 100,000 Pa, held constant.
Volume change: An increase of 0.002 m^3.
Heat input: 300 J enters the gas.

A. No change because pressure is constant

B. An increase of 500 J

C. An increase of 100 J

D. A decrease of 300 J

Correct: C

4. (2 points) How much heat must this engine reject?

Unit: Energy direction and the second law.
Worked-case data: Heat input: An engine receives 500 J. Useful work: It delivers 150 J. Cycle: The working substance returns to its initial state.
Separate investigation observations:
Heat input: An engine receives 500 J.
Useful work: It delivers 150 J.
Cycle: The working substance returns to its initial state.

A. 650 J

B. 350 J

C. 500 J

D. 0 J

Correct: B

5. (2 points) Joule’s work connected…

Unit: Historical experiments and modern thermodynamic reasoning.
Worked-case data: A mechanical heater adds 1000 J to a system with total capacity 500 J/K, neglecting losses. Predict delta T.

A. Only rock sorting

B. Only salinity and light

C. Mechanical work and thermal energy

D. Only genetics and proteins

Correct: C

6. (2 points) Q=mcdelta T gives…

Unit: Conduction, convection, radiation, and heat capacity.
Worked-case data: Heat 0.20 kg water by 10 K with c=4180 J/(kg K), neglecting losses.

A. Energy

B. Power without time

C. A universal cooling rate

D. Pressure

Correct: A

7. (2 points) Doubling mass at the same path doubles…

Unit: Latent heat, heating curves, and calorimetry.
Worked-case data: Melt 0.10 kg ice at 0 °C using Lf=334 kJ/kg. Find energy.

A. Kelvin offset

B. Required energy

C. Transition temperature necessarily

D. Specific heat

Correct: B

8. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Gas processes with graphs and simple algebra.
Worked-case data: Pressure: 100,000 Pa, held constant. Volume change: An increase of 0.002 m^3. Heat input: 300 J enters the gas.
Separate investigation observations:
Pressure: 100,000 Pa, held constant.
Volume change: An increase of 0.002 m^3.
Heat input: 300 J enters the gas.

answer: Calculate work: 100,000 x 0.002 = 200 J. Use change in internal energy = heat added - work done. Subtract 300 - 200 = 100 J. (4 points).

9. (2 points) For one cycle, W equals…

Unit: Heat engines, Carnot cycles, refrigerators, and heat pumps.
Worked-case data: For Th=600 K and Tc=300 K, find Carnot efficiency and refrigerator COP.

A. Qc-Qh for an engine output

B. Qh-Qc

C. Qh+Qc always

D. Zero necessarily

Correct: B

10. (2 points) Gas-law ratios require…

Unit: Systems, temperature, and thermal equilibrium.
Worked-case data: Convert 27 °C to Kelvin and compare a 10 °C temperature change.

A. Kelvin

B. Celsius

C. Fahrenheit

D. An arbitrary zero

Correct: A

11. (4 points) Why is pressure important?

Unit: Phases, diagrams, and transition paths.
Worked-case data: During ideal melting at 0 °C, added energy increases what?

answer: It changes equilibrium transition conditions. (4 points).

12. (2 points) Equal T means equal average translational…

Unit: Ideal gases and kinetic molecular theory.
Worked-case data: For one mole at 300 K in 0.010 m^3, estimate pressure using R=8.314 J/(mol K).

A. Density

B. Molar mass

C. Energy

D. Speed for all masses

Correct: C

13. (2 points) Adiabatic expansion with positive Wby gives delta U…

Unit: Internal energy and the first law.
Worked-case data: A system receives 500 J heat and does 200 J work. Find delta U and chemistry work.

A. Positive

B. Zero always

C. Undefined always

D. Negative

Correct: D

14. (4 points) Why can real efficiency be lower?

Unit: Heat engines, Carnot cycles, refrigerators, and heat pumps.
Worked-case data: For Th=600 K and Tc=300 K, find Carnot efficiency and refrigerator COP.

answer: Irreversibility and practical transfer and operating limitations reduce performance. (4 points).

15. (4 points) Why define a boundary?

Unit: Systems, temperature, and thermal equilibrium.
Worked-case data: Convert 27 °C to Kelvin and compare a 10 °C temperature change.

answer: It determines which matter and energy transfers count. (4 points).

16. (4 points) Why distinguish time and energy axes?

Unit: Phases, diagrams, and transition paths.
Worked-case data: During ideal melting at 0 °C, added energy increases what?

answer: Time requires knowledge of net heating power to infer energy. (4 points).

17. (4 points) Why can Cp exceed Cv?

Unit: Ideal gases and kinetic molecular theory.
Worked-case data: For one mole at 300 K in 0.010 m^3, estimate pressure using R=8.314 J/(mol K).

answer: At constant pressure, heating includes expansion work. (4 points).

18. (4 points) Translate Wby=100 J to won.

answer: won=-100 J. (4 points).